

$$F(A, B, C, D) = \Sigma m(0, 6, 7, 12, 14) + \Sigma d(9, 15)$$

NEW FUNCTION

2. Finding Prime Implicants

Group	Decimal	Binary
0	0	0000 ✱
1	6	0110 ✓
1	12	1100 ✓
1	9	1001 ✱
2	7	0111 ✓
2	14	1110 ✓
3	15	1111 ✓

3. Finding Prime Implicants

Group	Decimal	Binary
0	14, 6	-110 ✓
0	6, 7	011- ✓

Group	Decimal	Binary
0	12, 14	11-0 ✖
1	15, 7	-111 ✓
1	14, 15	111- ✓

4. Finding Prime Implicants

Group	Decimal	Binary
0	14, 15, 6, 7	-11- ✖

The Prime Implicants Are
A'B'C'D', AB'C'D, ABD', BC

6. Prime Implicants Chart

Minterms	0	12	14	6	7	Cost
A'B'C'D'	X					5
ABD'		X	X			4
BC			X	X	X	3

7. Finding Unique Minterms

Minterms	0	12	14	6	7	Cost
$A'B'C'D'$	X					5
ABD'		X	X			4
BC			X	X	X	3

The Essential Implicants Are
 $A'B'C'D'$, ABD' , BC

Possible Function Minimizations
 $F = A'B'C'D' + ABD' + BC$